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DEFINITION

Microsoft Azure Key Vault

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What is Microsoft Azure Key Vault?

Microsoft Azure Key Vault is a cloud-based security service offered by Microsoft as part of its Azure platform. It provides a secure and centralized storage solution for cryptographic keys and secrets, such as passwords, certificates and keys used for encryption.

Azure Key Vault allows organizations to securely store and manage sensitive information, such as application secrets and cryptographic keys, in a way that is easily accessible to authorized users and applications. This allows organizations to enhance their IT security and reduce the risk of [data breaches](#).

Neither applications nor Microsoft have direct access to keys. Instead, users grant permissions for their own and third-party applications to use the keys as needed.

1 Permissions 2 Principal 3 Application (optional) 4 Review + create

Configure from a template

Key, Secret, & Certificate Management

Key permissions	Secret permissions	Certificate permissions
Key Management Operations ☒ Select all ☒ Get ☒ List ☒ Update ☒ Create ☒ Import ☒ Delete ☒ Recover ☒ Backup ☒ Restore Cryptographic Operations ☐ Select all ☐ Decrypt ☐ Encrypt ☐ Unwrap Key ☐ Wrap Key ☐ Verify ☐ Sign	Secret Management Operations ☒ Select all ☒ Get ☒ List ☒ Set ☒ Delete ☒ Recover ☒ Backup ☒ Restore Privileged Secret Operations ☐ Select all ☐ Purge	Certificate Management Operations ☒ Select all ☒ Get ☒ List ☒ Update ☒ Create ☒ Import ☒ Delete ☒ Recover ☒ Backup ☒ Restore ☒ Manage Contacts ☒ Manage Certificate Authorities ☒ Get Certificate Authorities ☒ List Certificate Authorities ☒ Set Certificate Authorities ☒ Delete Certificate Authorities Privileged Certificate Operations ☐ Select all



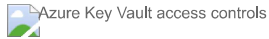
Edit permissions to control access Azure Key Vault.

What key management scenarios can Azure Key Vault support?

Azure Key Vault supports several [key management](#) scenarios:

- **Generation.** Azure Key Vault can generate and manage cryptographic keys.
- **Storage.** Azure Key Vault can store cryptographic keys and secrets, such as passwords and certificates, in a secure and centralized location.
- **Encryption.** Data can be encrypted and decrypted using the cryptographic keys stored in the service.
- **Management.** Azure Key Vault provides a centralized and auditable way to manage keys and secrets, including the ability to set access policies, track usage and rotate keys.

Azure Key Vault can also be integrated with other Azure services, such as Azure Virtual Machines, [Azure Functions](#) and [Azure DevOps](#), to provide secure and seamless access to cryptographic keys and secrets.



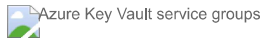
Apply access controls to a key vault with Azure Key Vault.

What are the benefits of using Azure Key Vault for key management?

There are several benefits to organizations that use Microsoft Azure Key Vault to help manage their keys and passwords:

- **Safe centralized storage of keys.** Azure Key Vault provides a secure and centralized location to store sensitive information, such as passwords, certificates and encryption keys. This secure storage reduces the risk of data breaches and maintains the security of systems and data.
- **Key management.** Azure Key Vault provides a centralized and auditable way to manage cryptographic keys and secrets, including the ability to set access policies, track usage and rotate keys.
- **Added protection for other Azure services.** Azure Key Vault can be integrated with other Azure services allowing organizations to further reduce the risk of data breaches and improve the security of their cloud-based resources.
- **Compliance with regulations.** Azure Key Vault helps organizations meet their compliance requirements by providing a secure and auditable way to store and manage cryptographic keys and secrets. This helps organizations demonstrate

compliance with regulations such as the [Payment Card Industry Data Security Standard](#) while reducing the risk of penalties for noncompliance.



With Azure Key Vault, admins can group key vaults by services.

What does Microsoft Azure Key Vault cost?

Microsoft Azure Key Vault's pricing structure is based on the number of requests made to the service and the storage of keys and secrets. The pricing is divided into two components:

- **Requests.** Azure Key Vault's price is based on the number of requests made to the service, such as creating and accessing keys, secrets and certificates. Requests are charged on a per-operation basis, and the price varies depending on the volume of requests made.
- **Storage.** Azure Key Vault is also priced based on the amount of storage used for keys and secrets. The price for storage varies based on the volume of stored data.

Azure Key Vault also offers a free tier that includes a limited number of requests and storage each month. This free tier lets organizations try Azure Key Vault before committing to a paid subscription.

What is Microsoft Azure? An Introduction



Learn how to best [protect data with these Azure Key Vault best practices](#).

This was last updated in April 2023

➤ Continue Reading About Microsoft Azure Key Vault

- [How to perform and automate key rotation in Azure Key Vault](#)
- [Compare Azure Key Vault vs. Kubernetes Secrets](#)
- [Using PowerShell for Azure service principal authentication](#)
- [Get to know the Azure resource hierarchy](#)
- [Deploy and manage Azure Key Vault with Terraform](#)

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